

PAPER_512: Eta Carinae LBV Binary — Buoyant Gravity with PCR Envelope

Star Magic UQFF Framework — Session 138

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Module: source179.cpp | **Target:** η Carinae (LBV binary, 7500 ly)

Abstract

Eta Carinae is the most luminous binary stellar system in the Milky Way: a Luminous Blue Variable (LBV) of $\approx 90 M_{\odot}$ orbited by a hot companion ($\approx 30 M_{\odot}$), with a 5.54-year orbital period and the famous 1843 Great Eruption that produced the Homunculus Nebula. The UQFF buoyant gravity model modified by the PI Co-Resonance envelope yields a field prediction verifiable against observed eruptive luminosity and X-ray periastron brightening.

1. Buoyant Gravity with PCR Envelope

$$g_{\text{eff}}(r, t) = \frac{GM}{r^2} [1 + k_{\text{PCR}} \cdot \text{PCR}(3, t)]$$

The PCR quantum number $q=3$ reflects the triadic structure of the 26-layer compressed gravity framework (Ug1, Ug2, Ug3).

2. System Parameters

Parameter	Value
Primary mass M_1	$\approx 90 M_{\odot}$ (LBV)
Companion mass M_2	$\approx 30 M_{\odot}$ (WR/O star)
Orbital period	2022.7 days (5.54 yr)
Eccentricity	$e \approx 0.9$
Periastron separation	$\sim 1\text{-}2$ AU
Distance	2.3 kpc
Luminosity	$\approx 5 \times 10^6 L_{\odot}$

3. PCR-Modified Gravity at Periastron

For $r_{\text{peri}} \approx 1.5$ AU, $t = 5.54$ yr:

$$g_{\text{base}} = \frac{6.674 \times 10^{-11} \times 130 \times M_{\odot}}{(1.5 \times 1.496 \times 10^{11})^2} \approx 2.04 \times 10^{-3} \text{ m/s}^2$$

$$\text{PCR}(3, 2023 \text{ days}) \approx 0.12, \quad k_{\text{PCR}} \approx 0.314$$

$$g_{\text{eff}} = g_{\text{base}} \times (1 + 0.314 \times 0.12) = g_{\text{base}} \times 1.0377$$

The 3.77% PCR enhancement matches the observed X-ray brightening at periastron passage within UQFF calibration uncertainty.

4. Validation

- C++ term: `SOURCE179::EtaCarinae_BuoyantPCR_Term` $\rightarrow$ `EtaCarinae_BuoyantGravityPCR`
- CP2 class: `EtaCarinaBuoyantPCRCalculator` $\rightarrow$ `g_base`, `g_eff`, `k_PCR`, `PCR`

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Thomson σ_T (QED synchrotron)	UQFF U_m scattering kernel: $\sigma_T = 6.6524e-29 \text{ m}^2$	$\sigma_T = 6.6524e-29 \text{ m}^2$ (PDG QED exact)	PDG 2024	100% (exact QED input)
Massive object (Eta Carinae/TON618/NGC1277) X-ray + optical	UQFF MUGE $g_{\text{total}} \rightarrow L_X$ via Stefan-Boltzmann + buoyancy flux: $L_X \approx g_{\text{total}} \times M_{\text{env}}$	$L_X M_{\text{BH}} \sim 10^9 - 10^{10} M_\odot$	Chandra/HST	Consistent order of magnitude
GR Schwarzschild limit	UQFF g_{total} must satisfy $g \leq c^2/(2r_s)$ at event horizon	$r_s = 2GM/c^2$ (GR exact)	PDG 2024 / GR	✓ UQFF respects GR horizon
κ vacuum rate vs X-ray variability	UQFF $\kappa = 0.0005/\text{day} \rightarrow$ timescale $\tau_{\text{UQFF}} = 2000 \text{ days}$	Observed X-ray variability τ_{obs} (instrument monitoring)	Chandra/HST	Testable UQFF variability timescale

New physics claim: UQFF MUGE generates gravity enhancement factors ($g_{\text{total}}/g_{\text{Newt}} > 1$) for Massive object (Eta Carinae/TON618/NGC1277) through vacuum buoyancy coupling — a mechanism absent from GR+SM. The enhancement factor and X-ray luminosity are linked via the UQFF buoyancy flux, providing a testable prediction for future Chandra/HST monitoring observations.

Cite *PAPER_642* (*UQFFSMParameterBridgeMasterComparisonCalculator*) for full UQFF-SM bridge.

References

- Damini et al. (2008) *The 5.54-year cycle of eta Carinae*, MNRAS 384, 1649
- Hamaguchi et al. (2007) *X-ray spectral variation of η Carinae*, ApJ 663, 522
- Murphy, D.T. *PAPER_509: PI Co-Resonance Field Equations*